ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 36: Pandas Plotting (Data Visualisation)

Aim

To visualise data using **Pandas built-in plotting functions** by generating a **histogram and bar chart**.

Algorithm

1. Import the required libraries.
2. Create a Pandas DataFrame containing student names and marks.
3. Generate a histogram to show the distribution of marks.
4. Generate a bar chart to compare marks of students.
5. Display the plots.

Program

```
# Lab 36: Pandas Plotting (Data Visualisation)


import pandas as pd

import matplotlib.pyplot as plt


# Create DataFrame

df = pd.DataFrame({

   "name": ["A", "B", "C", "D", "E"],

   "marks": [78, 85, 92, 66, 74]

})


# Histogram

df["marks"].plot(

   kind="hist",

   title="Marks Distribution",

   bins=5

)
```

```python
plt.xlabel("Marks")
plt.show()

# Bar chart
df.plot(
    x="name",
    y="marks",
    kind="bar",
    title="Marks by Student",
    legend=False
)

plt.ylabel("Marks")
plt.show()
```

Sample Output

- A **histogram** showing the distribution of student marks.
- A **bar chart** showing marks obtained by each student.

Result

Thus, **Pandas built-in plotting functions were successfully used to generate a histogram and a bar chart for data visualisation**.